

JIE (JENNIFER) SHAO

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EDUCATION

University of California, Los Angeles (UCLA), CA, US

Sep 2022 – Dec 2023

Master of Engineering in Artificial Intelligence

University of Washington (UW), WA, US

Sep 2019 – Aug 2022

B.A. in Mathematics

Relevant Coursework: Real Analysis, Probability I and II, Linear Analysis, Matrix Algebra, Scientific Computing, Database Systems, Data Visualization

RESEARCH INTERESTS

Data-centric AI and ML data systems: label noise and annotation quality in human-AI data pipelines, human-AI collaboration, and model evaluation.

RESEARCH EXPERIENCE

Shanghai Jiao Tong University, Research Assistant, Human-AI Annotation and Data Quality

Apr 2026 – Present

- Suggestion-Conditioned Label Noise (SCLN):** Defined a new class of label noise arising in human-AI data pipelines with model-in-the-loop annotation, in which the error depends jointly on the image, the AI suggestion, and how that suggestion is displayed, a structure no existing class-conditional or instance-dependent noise model captures. Showed that human confirmation of an AI pre-label does not remove this noise, contrary to the standard industry assumption. Instantiated the definition for 2D object detection (SCLN-Det) and formalized the counterfactual question of whether an annotator would have erred without the suggestion.
- AILA-Bench Benchmark Design:** Designed and ran a randomized annotation study in which 370 annotators labeled 100,000 BDD100K images under three matched conditions (Human-only, AI-assisted, AI-assisted with confidence), with per-image pairing so the same object is labeled with and without an AI suggestion; recorded interaction traces and AI error types for 567,963 source-linked labels. Found that AI assistance barely lowers the final detection error rate (31.5% to 30.8%) while raising localization error from 3.9% to 13.9%, and that 72.5% of wrong AI suggestions survive uncorrected into the final labels.
- SCLNScore Audit Method:** Developed SCLNScore, a per-label risk score that ranks labels by how much the AI suggestion raised their error risk, estimated with two calibrated XGBoost models in a T-learner design with cross-fitting and annotator-level effects to separate AI-induced contamination from ordinary task difficulty. Outperformed confidence-based review at flagging AI-induced errors (AUPRC 0.472, a 2.32x gain at a 1% review budget) and transferred to nuImages (AUPRC 0.418 vs. 0.308 baseline).
- End-to-End Research Ownership:** Owned the full cycle as sole author: literature survey, experimental design, annotation-platform instrumentation and data pipeline, model training and evaluation, manuscript writing, and public demo release. Paper accepted at WISE 2026 (Springer LNCS); preliminary version at the ECCV 2026 CDEL workshop.

PUBLICATIONS

- J. Shao. "Can Model-in-the-Loop Annotation Be Safely Scaled? Auditing Suggestion-Conditioned Label Noise in Human-AI Data Pipelines." 27th International Conference on Web Information Systems Engineering (WISE 2026), Venice, Italy, Nov 2026. Springer LNCS, to appear. Preliminary version: "SCLN-Det: Auditing Suggestion-Conditioned Label Noise in Model-in-the-Loop Object Detection Annotation," Second Workshop on Curated Data for Efficient Learning (CDEL), ECCV 2026, Malmö, Sep 2026 (non-archival), openreview.net/forum?id=FlV8ZCa3VK
- J. Shao, Z. Ouyang, and Y. Zeng. "PoW and PoS and Related Applications." 2021 International Conference on Electronic Information Engineering and Computer Science (EIECS), pp. 59-62. doi:10.1109/EIECS53707.2021.9588080

TECHNICAL SKILLS

Research Methods: randomized and paired experimental design, causal inference (potential outcomes, T-learner, cross-fitting), calibration and precision-recall evaluation, label-noise and annotation-quality auditing, benchmark construction (BDD100K, nuImages), LaTeX

ML / Vision: PyTorch, XGBoost, YOLOv8, Grounding DINO, SAM2, OpenCV, CUDA, GPU optimization, distributed training (DDP), vector databases

Engineering: Python, C++, Java, SQL, Bash, Kubernetes, Docker, Argo Workflows, FastAPI, Spark, Airflow, Kafka, AWS (S3, EC2, IAM), MongoDB, PostgreSQL, Redis, Prometheus, Grafana, CI/CD

INDUSTRY EXPERIENCE

Pony.ai, Software Engineer – Data Evaluation, Fremont, CA

May 2024 – Apr 2026

- Model-in-the-Loop Annotation Platform (SAM2):** Architected the SAM2 embedding and inference pipeline behind a production labeling platform, cutting manual annotation time by 60%+ and enabling annotators to label complex multi-camera scenes 3-5x faster; implemented distributed GPU inference on Argo Workflows, FastAPI model serving, per-camera temporal video-predictor state for 6-camera rigs, and MongoDB vector storage serving 100+ annotators across regions. Observed annotators accepting wrong model pre-labels at scale in this system, the production failure that motivated the SCLN research.
- Model Evaluation Framework and Metrics:** Designed C++ evaluation units for a production perception stack (3D detection, occupancy, drivable area, lane) executed in simulation replay across 200+ real-world scenarios; defined a severity-tiered taxonomy of critical misses, occlusion-split miss and false-positive metrics, per-class IoU/mIoU, and Chamfer-distance boundary matching. Overhauled detection matching with occupancy-tracked bipartite assignment, raising correct-classification counts by 24.6% and cutting total error rate by 15%. Built multi-dimensional metric slicing (weather, geography, distance band, occlusion, confidence) to localize long-tail regressions, and defined ground-truth coverage rules (no-label-zone masks, unlabeled-type exclusion, valid-range filtering) so metrics measure system performance rather than annotation gaps.
- Evaluation and Data Infrastructure:** Built ETL pipelines processing 5+ TB/day of multi-modal sensor data (camera, LiDAR, radar, GPS) with validation gates and cross-region coordinate transforms; ran continuous evaluation on Kubernetes and Argo with dynamic GPU scheduling, sustaining 10,000+ daily batch jobs across 50+ GPU nodes at 99% reliability; led a zero-downtime InfluxDB to MongoDB migration of the metrics platform with 50-70% query-latency reduction, and cut annual infrastructure spend by \$200K+ through tiered storage and GPU job packing.

ByteDance Ltd., Data Scientist Intern, Guangzhou, CN

Jun 2021 – Dec 2021

- Creator Analytics Pipelines:** Built Airflow data pipelines and automated Tableau and D3.js dashboards for internal stakeholders, contributing to a 1.67x increase in gross merchandise value of company-produced videos; developed an ETL/Cloudera pipeline over 15TB+ of TikTok video-session and user-behavior data that cut processing time for ML and risk analysis by 16.5%.

PROJECTS

Autonomous Driver Simulation (Python, C++) | AI and Deep Learning Capstone, UCLA

Summer 2023

- Migrated the perception model from CNN to R-CNN in PyTorch and implemented distributed training with DistributedDataParallel, reducing training time by 12% while classifying 2,000+ region proposals per image within 47 ms; built OpenCV lane detection on front-camera video with Gaussian smoothing, Canny edges, Hough transform, and perspective transformation.